

Philosophy of Space and Time

Lecture 4: *Reichenbach; Geometry and Physics*

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Today's Plan

- 1 Recap: Poincaré's Conventionalism
- 2 Reichenbach and Coordinative Definitions
- 3 Assessment
- 4 Transition to Newton

Where We Left Off

Poincaré argued that no observation can determine the geometry of space *in isolation*:

Observed result = Geometry G + Physical theory P

- The heated disk: same measurements, two $G + P$ “packages”
- The choice of G is a **convention**: not a priori, not empirical
- Guided by simplicity, other pragmatic considerations, not truth

Today: Reichenbach's more precise account—and what we can learn from this line of thought.

Reichenbach

Reichenbach's Starting Point

“Mathematics reveals the possible spaces; physics decides which among them corresponds to physical space... But what methods should physics employ?”

— Reichenbach, *Philosophy of Space and Time* (1928)

Two extremes Reichenbach wants to avoid (for reasons similar to Poincaré's):

- Kant: geometry is a priori and necessarily Euclidean
- Naïve empiricism: just measure and see

The Problem of Meaning

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Until you *define* a Smoot, the claim is meaningless

(A Smoot $\approx$ 1.70 m — the height of Oliver Smoot, MIT class of 1962. The Harvard Bridge is 364.4 Smoots long.)

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Similarly: “space is Euclidean” is meaningless until we specify what physical objects correspond to “distance,” “straight line,” etc.

Coordinative Definitions

Key Concept

A **coordinative definition** links an abstract mathematical concept to a physical object (or procedure).

- Example: “one meter” = the length of a particular rod in Paris
- This is a *stipulation*, not a factual claim
- You cannot “check” whether the meter rod is really one meter
- Without such definitions, abstract geometry has no empirical content

Norton (C2) discusses how Reichenbach developed this idea; compare with Poincaré’s argument in Huggett Ch. 13.

Coordinative Definitions for Geometry

To make geometry empirical, we must coordinate:

- “Distance” $\leftrightarrow$ a measuring rod
- “Straight line” $\leftrightarrow$ a light ray (or something else, e.g. taut string)
- “Congruent” $\leftrightarrow$ rigid body transport

A crucial stipulation: the rod does not change length during transport.

?? *But how would we detect a change, if every rod changes the same way?*

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Reichenbach’s solution: define rigid bodies as those not affected by **differential** forces, and handle “universal” forces by stipulation.

Universal vs. Differential Forces

Differential Forces

- Affect materials differently
- Detectable and correctable
- Example: thermal expansion (brass vs. steel)
- Can be “shielded against”

Universal Forces

- (As in Poincaré's disk)
- Affect *all* bodies identically
- Undetectable by any comparison
- No shielding possible

Reichenbach's Stipulation

Define rigid bodies as those not affected by differential forces (allows corrections for known ones). **Set universal forces to zero by definition.**

Logical vs. Technical Impossibility

Important distinction:

- **Technical impossibility:** we can't count all the molecules in this room, but there is a definite number
- **Logical impossibility:** we can't measure the meter rod to check if it is one meter. (The question presupposes the definition.)

The need for a coordinative definition is not a practical limitation—it is a **logical** one. Without a definition of “one meter,” the question “how many meters long is this?” has no answer.

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Similarly: without a definition of “rigid,” the question “is this rod the same length as before?” has no answer.

The Relativity of Geometry

Theorem $\ominus$

Given any geometry G' to which instruments conform, we can always find a universal force F such that the “true” geometry is any other geometry G .

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- Geometry is **relative** to a choice of coordinative definitions
- But *once the definitions are fixed*, the geometry is objectively determined

Convention and Objectivity

Reichenbach insists the result is *not* a slide into subjectivism:

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Similarly: geometry is relative to coordinative definitions—but given the definitions, geometry is objective.

Comparison with Kant: Relativized A Priori

Reichenbach's coordinative definitions share something with Kant's view of geometry as synthetic a priori:

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- Reichenbach: coordinative definitions are constitutive but **revisable** and **arbitrary**

“The contribution of reason is not expressed by the fact that the system of coordination contains unchanging elements, but by the fact that arbitrary elements occur in it.”

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A “relativized a priori”: constitutive principles that can change with new theories.

Poincaré and Reichenbach

	Poincaré	Reichenbach
Scope	Geometry only	Potentially broader
Key concepts	Rigid body motions	Coordinative definitions
Convention	Choice among constant-curvature geometries	Choice of coordinative definitions (universal forces)
Preferred geometry	Euclidean (simplest)	Whichever sets universal forces to zero
Comparison to Kant	Partly sympathetic	"Relativized a priori"

Break

Assessment

Is Conventionalism the Final Word?

Several worries:

1. **Triviality:** is this just relabeling? Or does it reveal something deep?
2. **Simplicity:** if Euclidean + bizarre forces gives the same predictions as non-Euclidean + no forces, is the simpler description merely “convenient”? If it is *better*, in what sense?
3. **Scope:** Poincaré restricted conventionalism to geometry: is that principled, or arbitrary?

Beyond Conventionalism: Definitions as Discovery

The conventionalists were right about something:

- Physical geometry depends on **constitutive principles** (or coordinative definitions) that assign physical meaning to geometric concepts
- These principles are not ordinary empirical claims, more like definitions

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But were they right to call these principles *conventions*?

- An alternative: such principles arise from **conceptual analysis**: working out what is *implicit* in established reasoning
- “Reflective Equilibrium”: revise concepts so that they fit appropriately with how they have been successfully applied (e.g., “force” and “mass”)
- Constitutive $\neq$ conventional: a principle can define a concept *and* result from reflective equilibrium

The Deeper Lesson

Regardless of where one lands on the conventionalism debate:

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This insight—that the structure of space and time is *entangled with dynamics*—will be central as we now turn to debates in the seventeenth century.

Transition: Space, Time, Motion

From Geometry to Motion

So far we have asked: *what is the geometry of space?*

Now we shift to a related but distinct question:

?? *What is the nature of space and time? And what structures must they have for physics (in particular, for a theory of motion) to work?*

This is the question that drove the great 17th-century debates.

The Problem of Motion

Motion = change of position over time.

- Position *with respect to what?*
- How do we distinguish a body that is “truly” moving from one that merely *appears* to move?
- Does it even make sense to ask whether a body is “really” in motion?

Two broad options:

1. **Absolute**: motion defined with respect to space itself
2. **Relational**: motion defined only relative to other bodies

What Does a Theory of Motion Need?

At a minimum, to state the laws of motion we need:

- A notion of **straight line** (for “uniform motion in a straight line”)
- A notion of **distance** (for measuring how far something has moved)
- A notion of **elapsed time** (for “uniform velocity”)
- A way to identify **the same place** at different times (for “change of position”)

?? *These are geometric requirements. The epistemology of geometry is directly relevant to the physics of motion.*

Newton's Insight

Newton recognized that the laws of motion *require* a distinction between:

Inertial Motion

- Uniform, straight-line motion
- No net force acting
- Newton's First Law

Accelerated Motion

- Change in speed or direction
- Net force required
- Newton's Second Law

This distinction *presupposes* a background structure: “straight line,” “uniform,” “same place over time.”

Newton argued that only **absolute space** can provide this structure.

Descartes as Newton's Foil

- Descartes defined motion as transfer from the vicinity of contiguous bodies, a **relational** definition
- Newton objected: this makes the laws of motion incoherent
 - › No definite velocity or direction
 - › No definite trajectory over time
 - › Cannot distinguish straight from curved motion

?? *If motion is defined only relative to neighboring bodies, how can you state a law that says objects move in straight lines? Straight with respect to what?*

Preview: The Debate



Over the next two weeks:

- Galilean Relativity
- Newton's arguments for absolute space (bucket, globes)
- Leibniz's relational critique (shift arguments)
- Basic question: how much spacetime structure does physics *really* need?

Summary

1. Reichenbach makes the conventional element precise via **coordinative definitions**—stipulations linking abstract concepts to physical objects
2. Geometry is **relative** to coordinative definitions, but objective once those are fixed
3. The conventionalists were right that geometry depends on **constitutive principles**—but those principles may arise from conceptual analysis rather than arbitrary stipulation
4. The entanglement of geometry with physics motivates our turn to Newton: what *structure* must space and time have for the laws of motion to make sense?

For Next Time

- **Read:** Newton, selections from “De Gravitatione” and the Scholium (C 4); Maudlin, Ch. 1 (M 1).
- **Think about:**
 - › Why does Newton think Descartes’ definition of motion is inadequate?
 - › What role does the bucket experiment play in Newton’s argument?

Tuesday: **Newton’s Analysis of Motion**

- The Scholium to the Definitions
- Absolute vs. relative space, time, and motion
- The bucket and the globes

Questions?

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